

Text Retrieval and Search Engines



Lecture 11 – Entity-Based Relevance Feedback

Based on work presented at the ICTIR conference

What is an Entity?

<DOCNO> FT921-7107</DOCNO>

It will allow astronomers to explore three quarters of the universe and study galaxies perhaps as far as 14bn light years away.

What is an Entity?

Entity name = Universe

Entity ID = 31880

$\Phi(\text{universe}) = 0.217$

- Φ : confidence score

<DOCNO> FT921-7107</DOCNO>

It will allow astronomers to explore three quarters of the universe and study galaxies perhaps as far as 14bn light years away.

Entity name = Research

Entity ID = 25524

$\Phi(\text{study}) = 0.085$

Universe

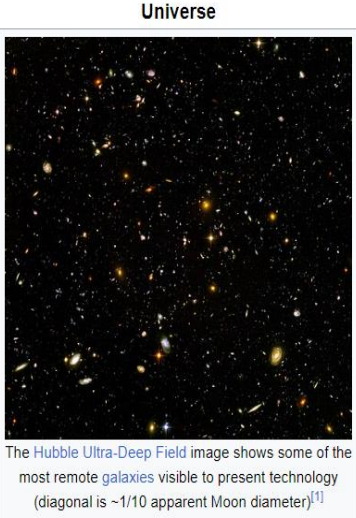
Article Talk

From Wikipedia, the free encyclopedia

For other uses, see [Universe \(disambiguation\)](#).

The **universe** is all of [space](#) and [time](#)^[a] and their contents,^[10] including [planets](#), [stars](#), [galaxies](#), and all other forms of [matter](#) and [energy](#). The [Big Bang](#) theory is the prevailing [cosmological](#) description of the development of the universe. According to this theory, space and time emerged together 13.787 ± 0.020 billion years ago,^[11] and the universe has been expanding ever since the Big Bang. While the spatial size of the entire universe is unknown,^[3] it is possible to measure the size of the [observable universe](#), which is approximately 93 billion [light-years](#) in diameter at the present day.

Some of the earliest [cosmological models](#) of the universe were developed by [ancient Greek](#) and [Indian philosophers](#) and were [geocentric](#), placing [Earth](#) at the center.^{[12][13]} Over the centuries, more precise astronomical observations led [Nicolaus Copernicus](#) to develop the [heliocentric model](#) with the [Sun](#) at the center of the [Solar System](#). In developing the [law of universal gravitation](#), [Isaac Newton](#) built upon Copernicus's work as well as [Johannes Kepler's](#) laws of planetary motion and observations by [Tycho Brahe](#).



Research

Article Talk

From Wikipedia, the free encyclopedia

This article is about the search for knowledge. For other uses, see [Research \(disambiguation\)](#).

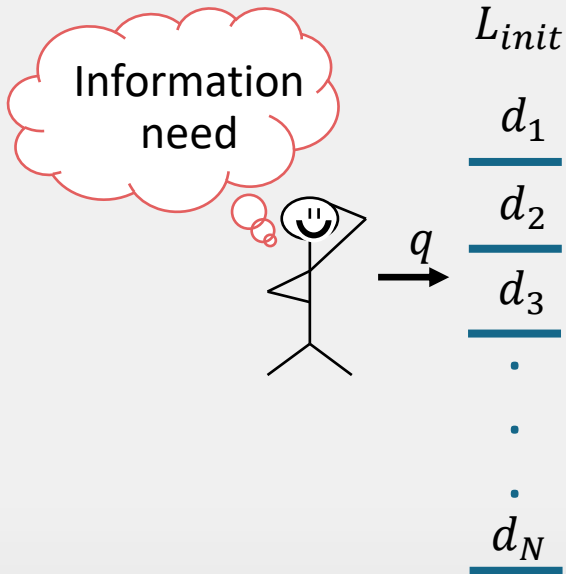
"[Researched](#)" redirects here. For the organisation, see [ResearchED](#).

"[Researcher](#)" redirects here. For other uses, see [Researcher \(disambiguation\)](#).

Research is "[creative](#) and systematic work undertaken to increase the stock of [knowledge](#)".^[1] It involves the collection, organization and analysis of evidence to increase understanding of a topic, characterized by a particular attentiveness to controlling sources of bias and error. These activities are characterized by accounting and controlling for biases. A research project may be an expansion on past work in the field. To test the validity of instruments, procedures, or experiments, research may replicate elements of prior projects or the project as a whole.

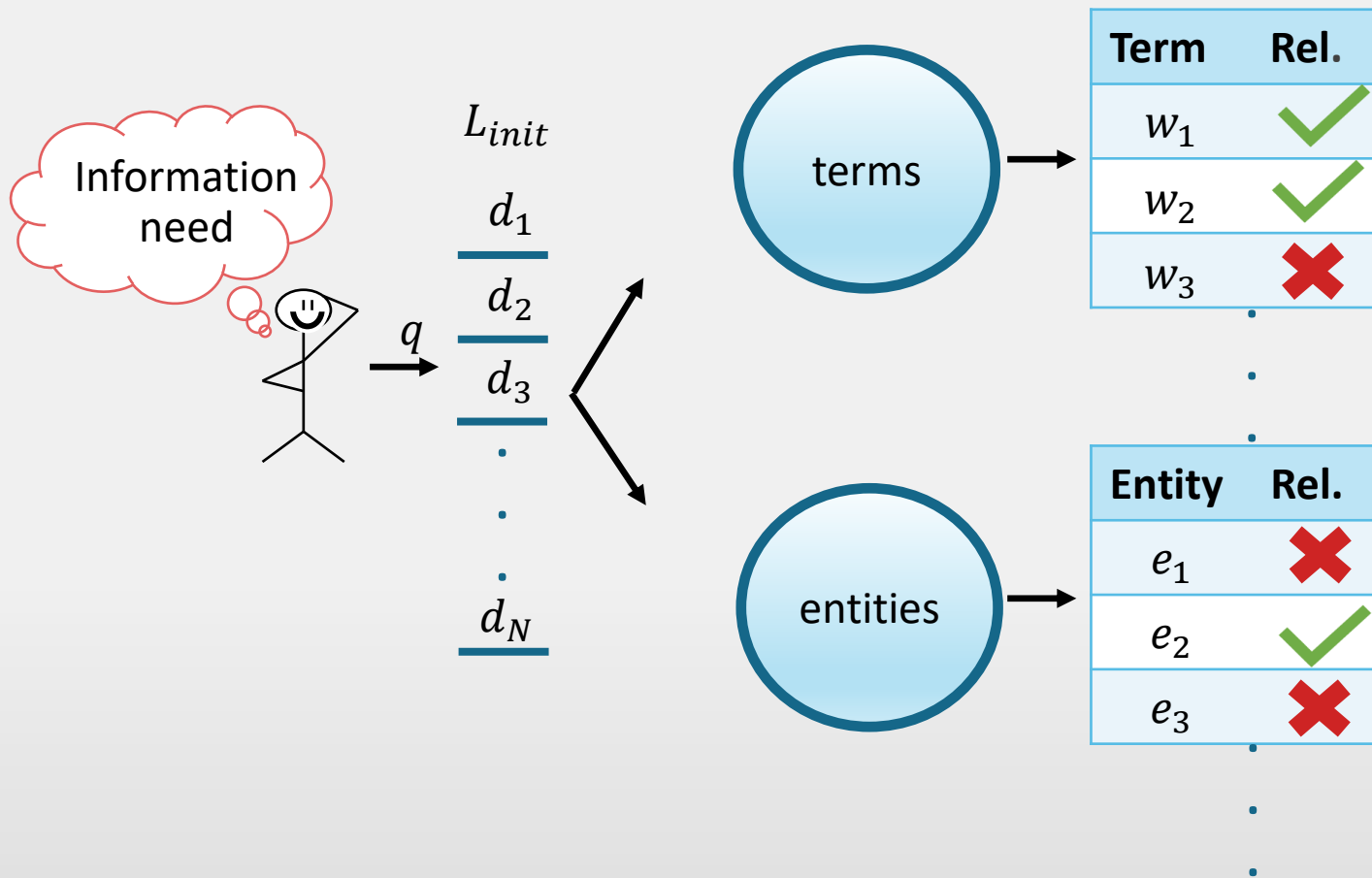
Our Relevance Feedback Framework

- L_{init} : initial list of top ranked N documents; q : query



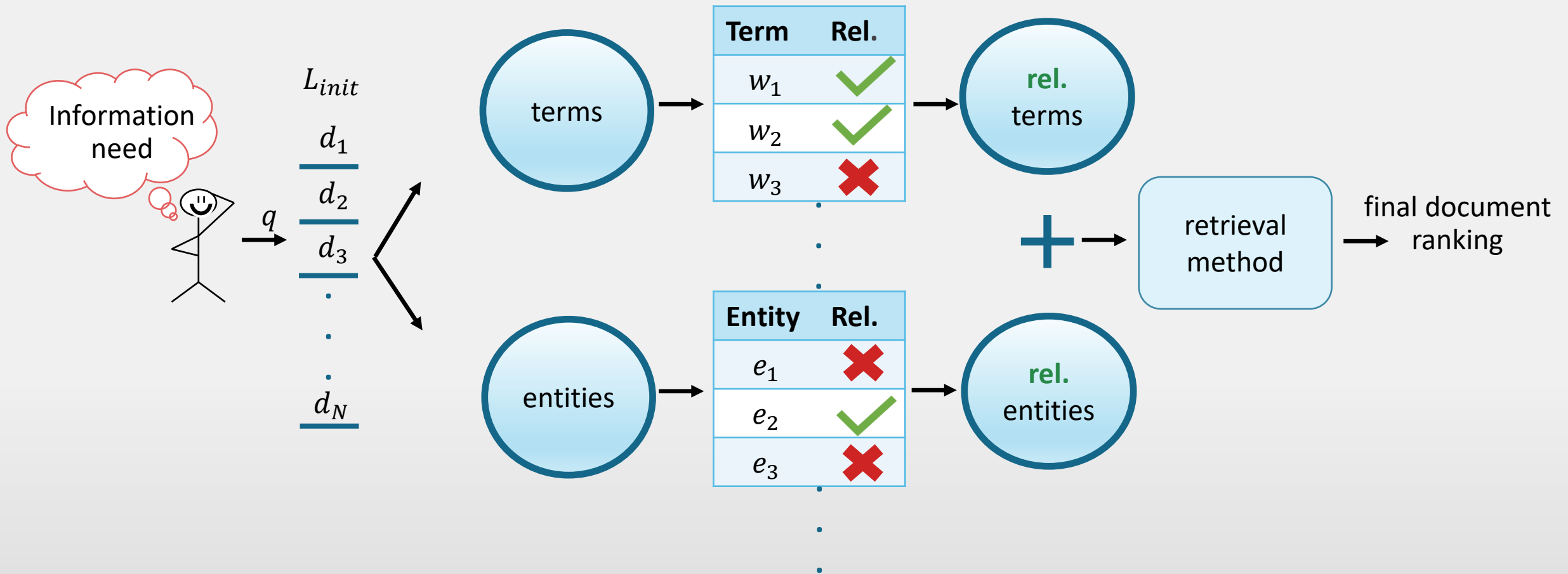
Our Relevance Feedback Framework

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Our Relevance Feedback Framework

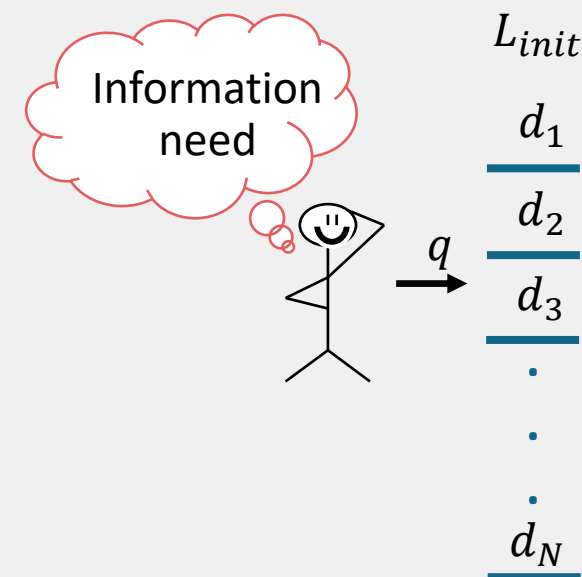
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STLM: Soft Confidence-Level Thresholding Language Model

[Raviv et al. '16]

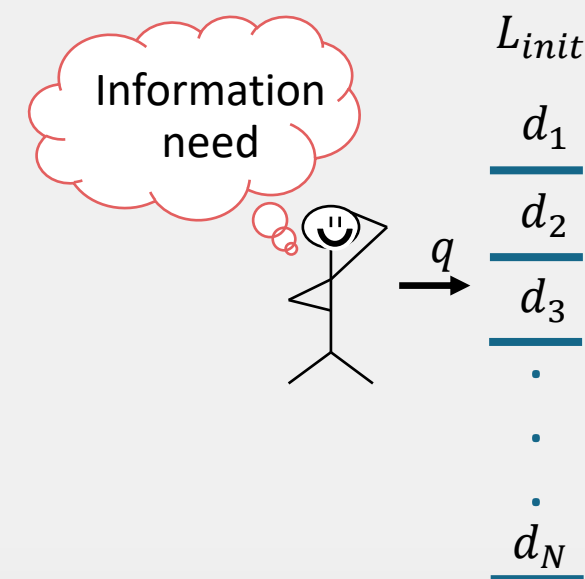
- STLM is used to produce L_{init}



STLM: Soft Confidence-Level Thresholding Language Model

[Raviv et al. '16]

- STLM is used to produce L_{init}
- $\mathcal{T}(\text{tokens}) \stackrel{\text{def}}{=} \mathcal{V}(\text{set of terms}) \cup \mathcal{E}(\text{set of entities})$



STLM: Soft Confidence-Level Thresholding Language Model

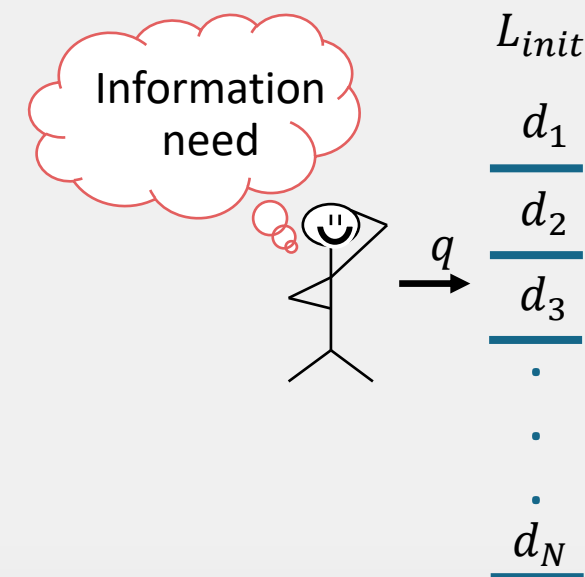
[Raviv et al. '16]

- STLM is used to produce L_{init}
- $\mathcal{T}(\text{tokens}) \stackrel{\text{def}}{=} \mathcal{V} \text{ (set of terms)} \cup \mathcal{E} \text{ (set of entities)}$

$$\bullet \quad pc(t, x) \stackrel{\text{def}}{=} \begin{cases} \lambda \, tf(t, x) & \text{if } t \in \mathcal{V} \\ (1 - \lambda) \sum_{m \in M(x): e(m)=t} \Phi(m) & \text{if } t \in \mathcal{E} \end{cases}$$

Annotations for the equation above:

- $pc(t, x)$ → pseudo count
- $tf(t, x)$ → term frequency of term t in text x
- $M(x)$ → set of all entity mentions in x
- $e(m)$ → entity mention in x
- $\Phi(m)$ → confidence score in mapping m to entity t



STLM: Soft Confidence-Level Thresholding Language Model

[Raviv et al. '16]

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pseudo count
↓
 $pc(t, x)$

term frequency of
term t in text x
↓
 $tf(t, x)$

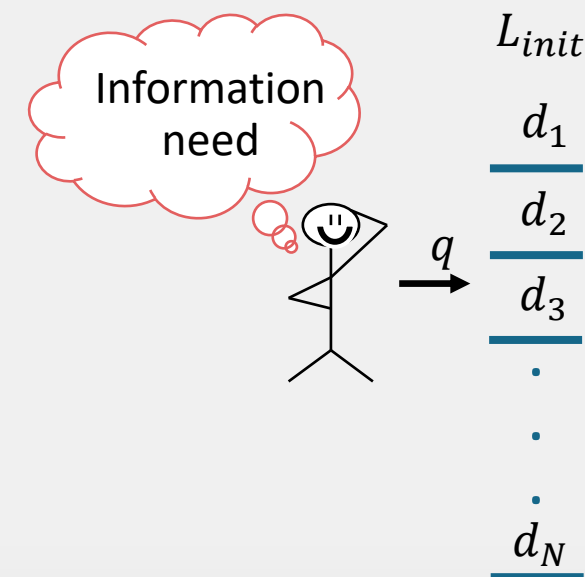
entity mention in x
↓
 $M(x)$

set of all entity
mentions in x
↓
 $M(x)$

confidence score in
mapping m to entity t
↓
 $\Phi(m)$

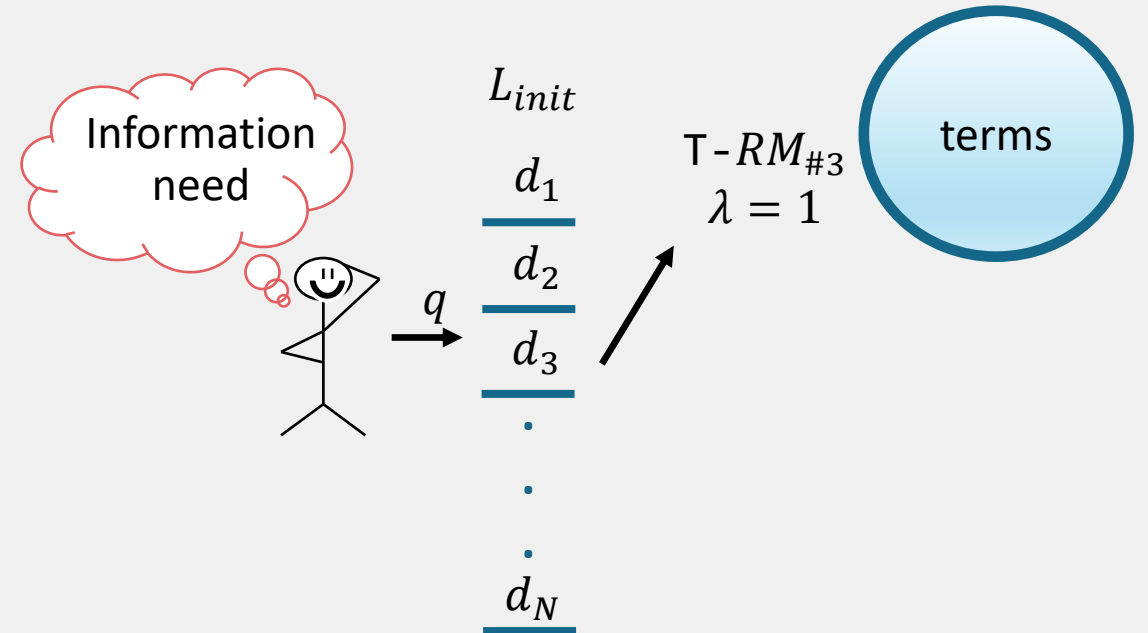
- $pl(x) \stackrel{\text{def}}{=} \sum_{t \in \mathcal{T}} pc(t, x)$

pseudo length
↓
 $pl(x)$



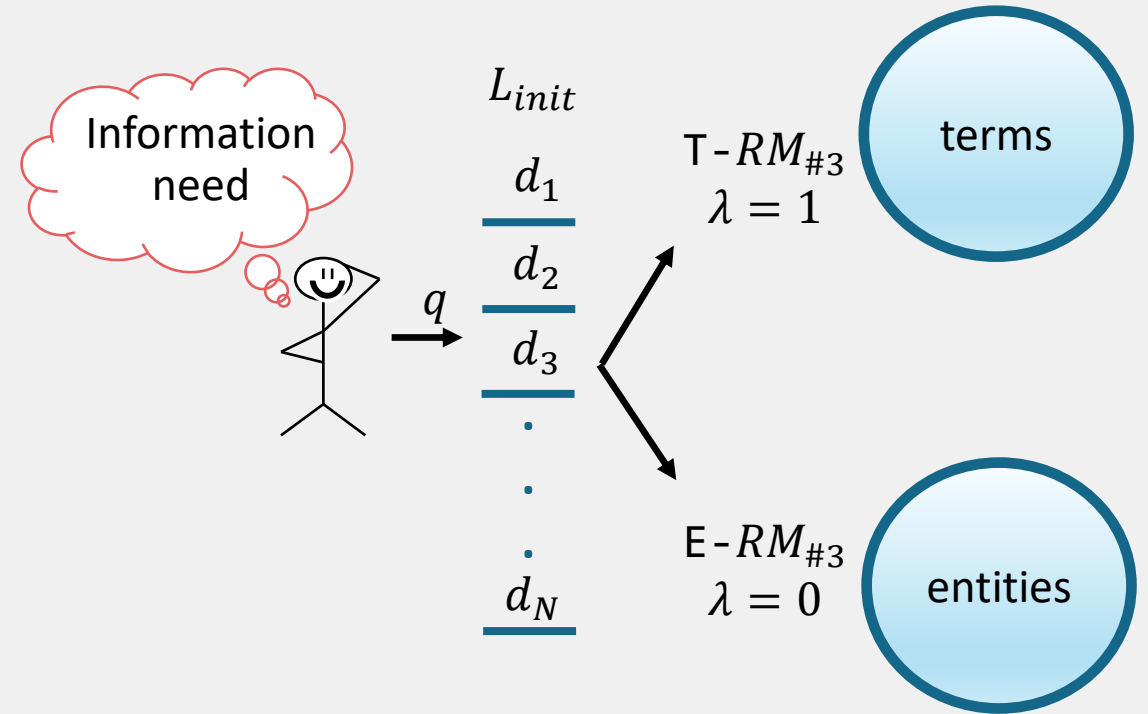
Token Selection

- $T-RM_{\#3}$: term-based relevance model
 - $\lambda = 1$
 - $pc(t, x) \stackrel{\text{def}}{=} tf(t, x)$



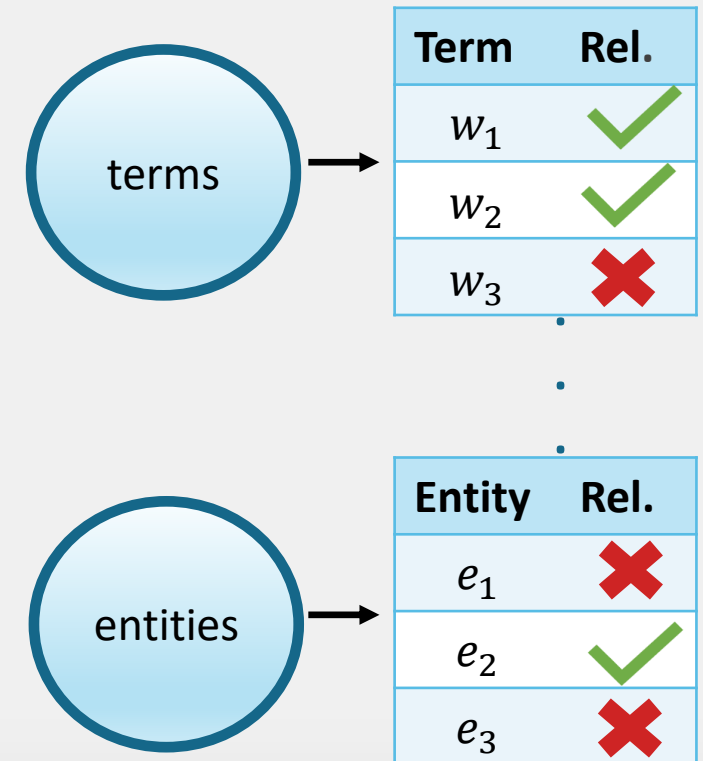
Token Selection

- $T-RM_{\#3}$: term-based relevance model
 - $\lambda = 1$
 - $pc(t, x) \stackrel{\text{def}}{=} tf(t, x)$
- $E-RM_{\#3}$: entity-based relevance model
 - $\lambda = 0$
 - $pc(t, x) \stackrel{\text{def}}{=} \sum_{m \in M(x): e(m)=t} \Phi(m)$



Presentation Approaches

- Terms
 - Term only [Kelly&Fu '06, Tan et al. '07]
 - Term + sentence from L_{init} [Kelly&Fu '06]
- Entities
 - Entity only
 - Entity + sentence from L_{init}
 - Entity + abstract from its Wikipedia page
- Entity linking: TagMe toolkit [Ferragina&Scaiella '10]
 - Data: Wikipedia dump from July 2014
- Human Intelligence Task (HIT) on MTurk platform
 - Majority vote: 3 votes per query-token pair
 - Quality check measures were applied



Example HIT

ROBUST query #607: “human genetic code”

Instructions

Short instructions

Select the Relevant button if an entity is **Relevant** to the given topic. Select the Not Relevant button if an entity is **Not Relevant** to the given topic.

A sentence which the entity appears in (**boldfaced** in parenthesis) is found to the right side of it.

Full instructions

Your task is to determine whether an entity is relevant to a given topic. The topic is described using a single paragraph. You are also provided with a query a user used to get information on the topic. Entities are titles of Wikipedia pages. A sentence which the entity appears in (**boldfaced** in parenthesis) is also provided.

What does it mean to be relevant?

If the entity and/or information about the entity would or should be included in a summary written about the topic then the entity is relevant to the topic. In other words, an entity is considered relevant if it is somehow related to the topic.

Are the entities relevant to the following topic?

Topic: What progress is being made in the effort to map and sequence the human genetic code?

Query: human genetic code

Entity1: Blood In another study, Loy found that blood (**Blood**) can survive on stone for much longer than 9,000 years, or even 50,000 years.

☐ Relevant

☐ Not Relevant

Entity2: Drug CVM accepts many AOAC and equivalent methods for use by laboratories in testing for drug residues (**Drug**) in animal tissues.

☐ Relevant

☐ Not Relevant

Entity3: Red River (Asia) New progress was made in irrigation works on the Huaihe, Honghe (**Red River (Asia)**), and Xinhe.

☐ Relevant

☐ Not Relevant

Entity4: DNA profiling A second trial involving DNA fingerprinting (**DNA profiling**) is expected to start next month in Orange County.

☐ Relevant

☐ Not Relevant

Entity5: Virus Olsen's vaccine does not contain the virus (**Virus**) itself, but proteins that are secreted by HTLV-1-infected animal cells that are grown in the laboratory.

☐ Relevant

☐ Not Relevant

The ETReF Dataset

- Entity and Term **R**elevance **F**eedback (ETReF) dataset
 - <https://github.com/Eilons/ETReF>

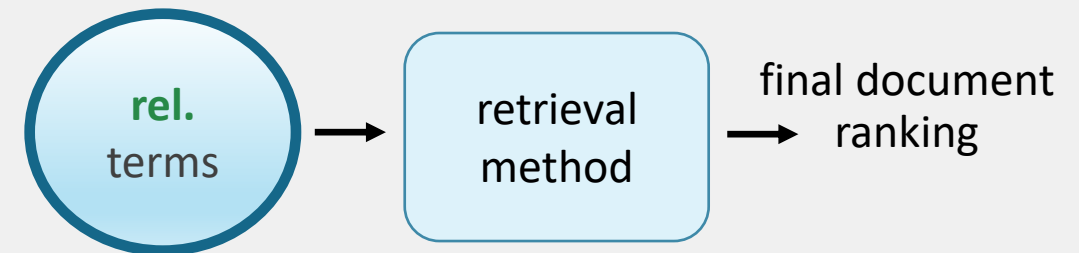
	Presentation	# tokens	% rel.	Example	
ROBUST	Term Only	12,740	39.0	<i>q</i> : “nobel prize winners” <i>w</i> : “award”	✓
				<i>q</i> : “nobel prize winners” <i>w</i> : “professor”	✗
	Entity Only	12,968	49.9	<i>q</i> : “nobel prize winners” <i>e</i> : “Particle physics”	✓
				<i>q</i> : “nobel prize winners” <i>e</i> : “Buckingham Palace”	✗
ClueWeb (09 Cat B)	Term Only	10,156	35.1	<i>q</i> : “poker tournaments” <i>w</i> : “holdem”	✓
				<i>q</i> : “poker tournaments” <i>w</i> : “texas”	✗
	Entity Only	10,352	30.6	<i>q</i> : “poker tournaments” <i>e</i> : “Texas hold'em”	✓
				<i>q</i> : “poker tournaments” <i>e</i> : “Poker strategy”	✗

Interpolated Token Feedback (ITF)

- $\mathcal{T}_r$: set of **relevant** tokens; $\delta_t = 1$ if token $t \in \mathcal{T}_r$ and 0 otherwise
- $p_q^{ITF}(t) \stackrel{\text{def}}{=} \beta \frac{pc(t,q)}{pl(q)} + (1 - \beta) \frac{\delta_t}{|\mathcal{T}_r|}$
 - β : free parameter

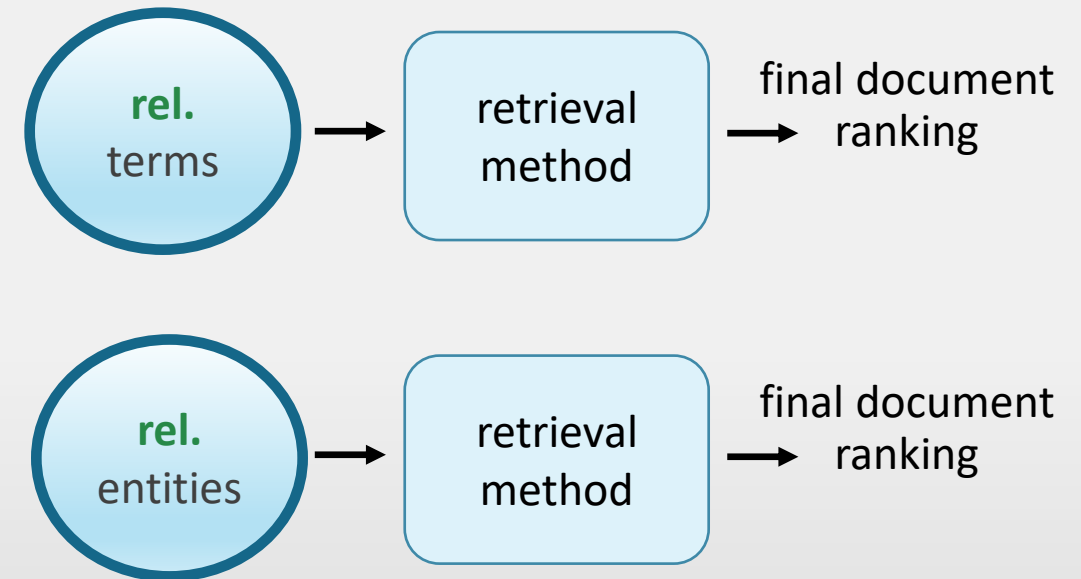
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 - β : free parameter
- T-ITF: term-based version
 - $\lambda = 1$
 - $\mathcal{T}_r$: set of **relevant** terms
 - $pc(t, x) \stackrel{\text{def}}{=} tf(t, x)$



Interpolated Token Feedback (ITF)

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- T-ITF: term-based version
 - $\lambda = 1$
 - $\mathcal{T}_r$: set of **relevant** terms
 - $pc(t, x) \stackrel{\text{def}}{=}} tf(t, x)$
- E-ITF: entity-based version
 - $\lambda = 0$
 - $\mathcal{T}_r$: set of **relevant** entities
 - $pc(t, x) \stackrel{\text{def}}{=} \sum_{m \in M(x): e(m)=t} \Phi(m)$

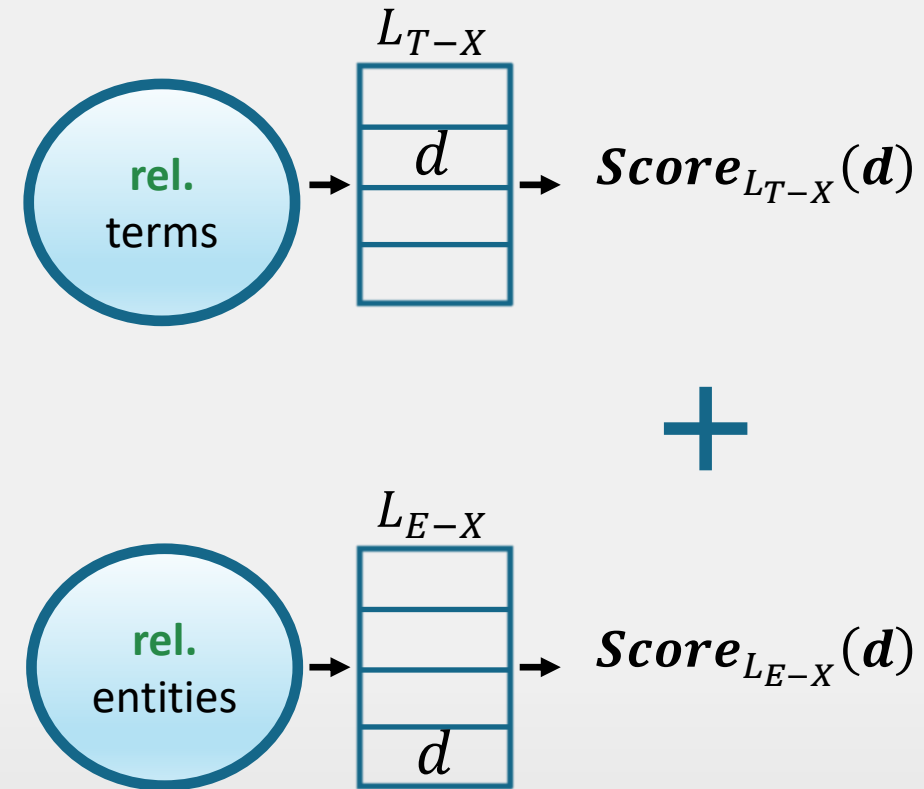


Reciprocal Rank Fusion

- Utilizing term and entity feedback together

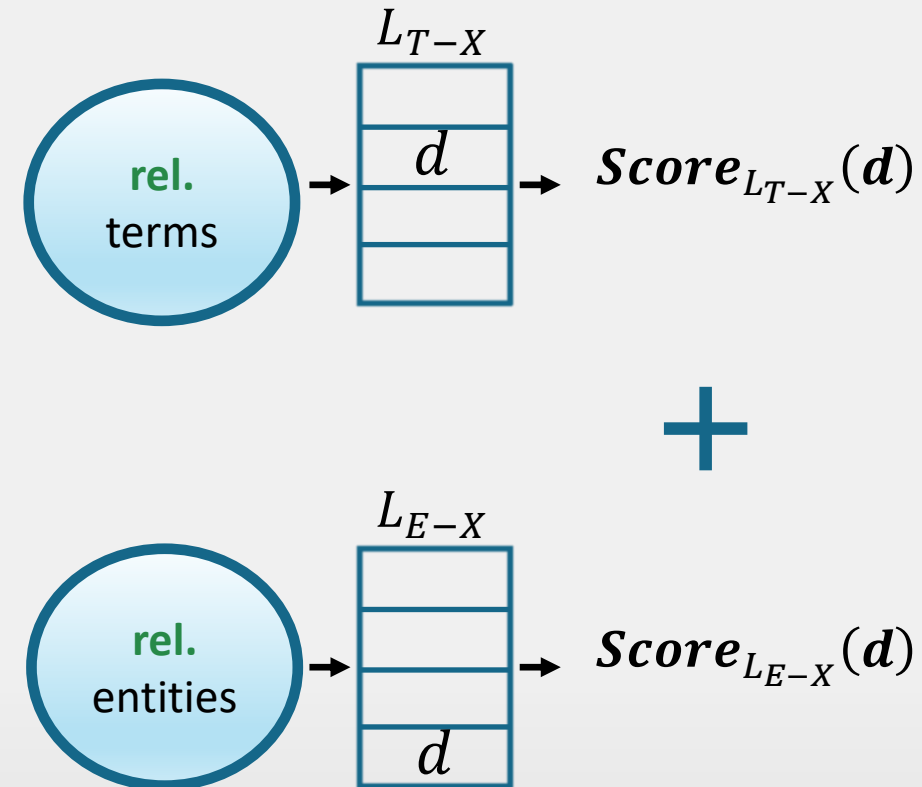
Reciprocal Rank Fusion

- Utilizing term and entity feedback together
- $F_X(q, d) \stackrel{\text{def}}{=} \lambda \text{score}_{L_{T-X}}(d) + (1 - \lambda) \text{score}_{L_{E-X}}(d)$
- $X: RM_{\#3}$ or ITF
 - E.g., **F-ITF**: reciprocal rank fusion of **T-ITF** and **E-ITF**

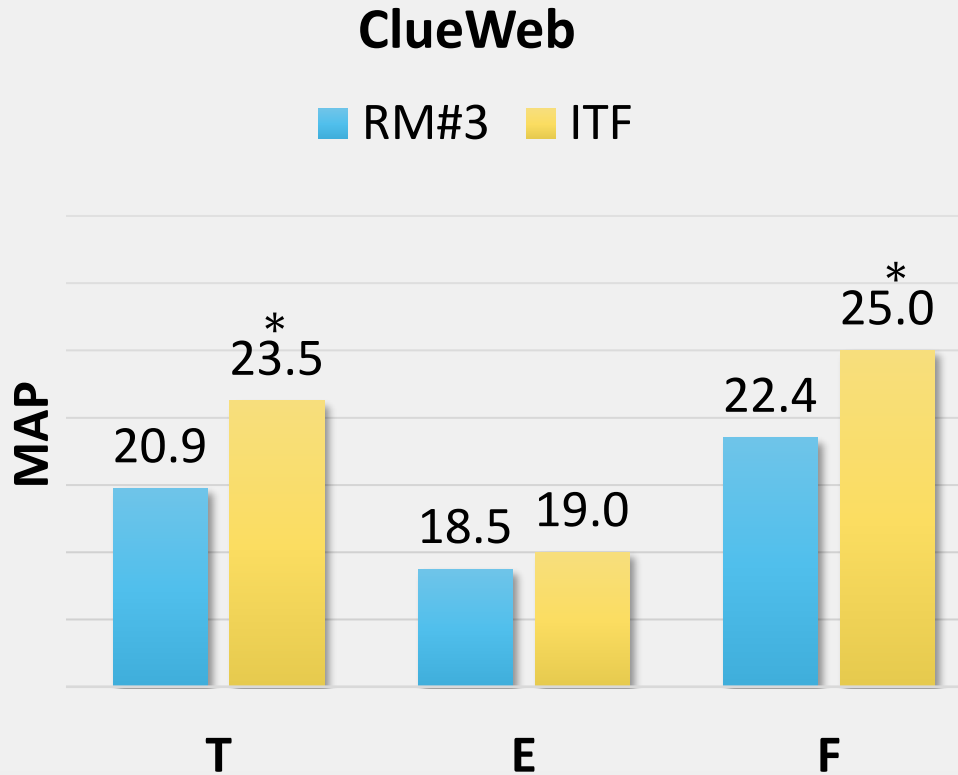


Reciprocal Rank Fusion

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- $F-X(q, d) \stackrel{\text{def}}{=} \lambda \text{score}_{L_{T-X}}(d) + (1 - \lambda) \text{score}_{L_{E-X}}(d)$
- $X: RM_{\#3}$ or ITF
 - E.g., **F-ITF**: reciprocal rank fusion of **T-ITF** and **E-ITF**
- Using a rank-to-score transformation [Cormack et al. '09]
 - $\text{Score}_L(d) \stackrel{\text{def}}{=} \frac{1}{v + r_L(d)}$
 - $r_L(d)$: d 's rank in list L
 - v : free parameter



Pseudo Vs. True Relevance Feedback



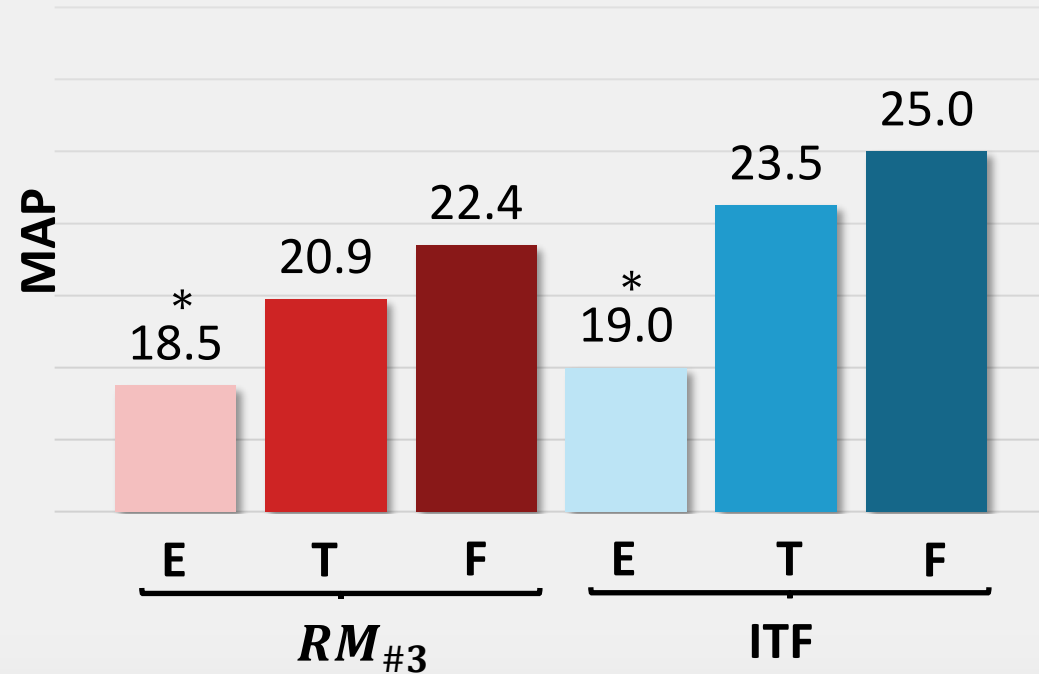
Findings:

1. True relevance feedback **outperforms** automatic query expansion
2. Term relevance feedback **outperforms** automatic term-based expansion
3. Entity relevance feedback is **comparable** to automatic entity-based expansion

*: statistically significant difference with $X\text{-}RM_{\#3}$ for $X \in \{T, E, F\}$

The Effect of the Feedback Token

ClueWeb



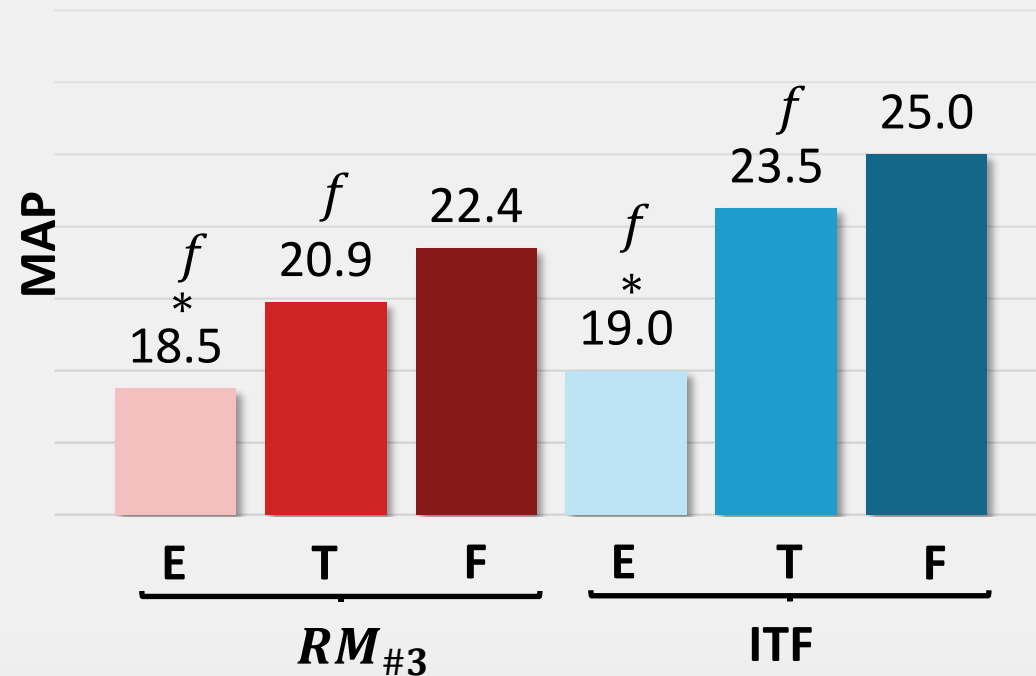
Findings

1. Entity relevance feedback is inferior to term feedback

‘*’: statistically significant difference with T-X for $X \in \{RM_{\#3}, ITF\}$

The Effect of the Feedback Token

ClueWeb



Findings

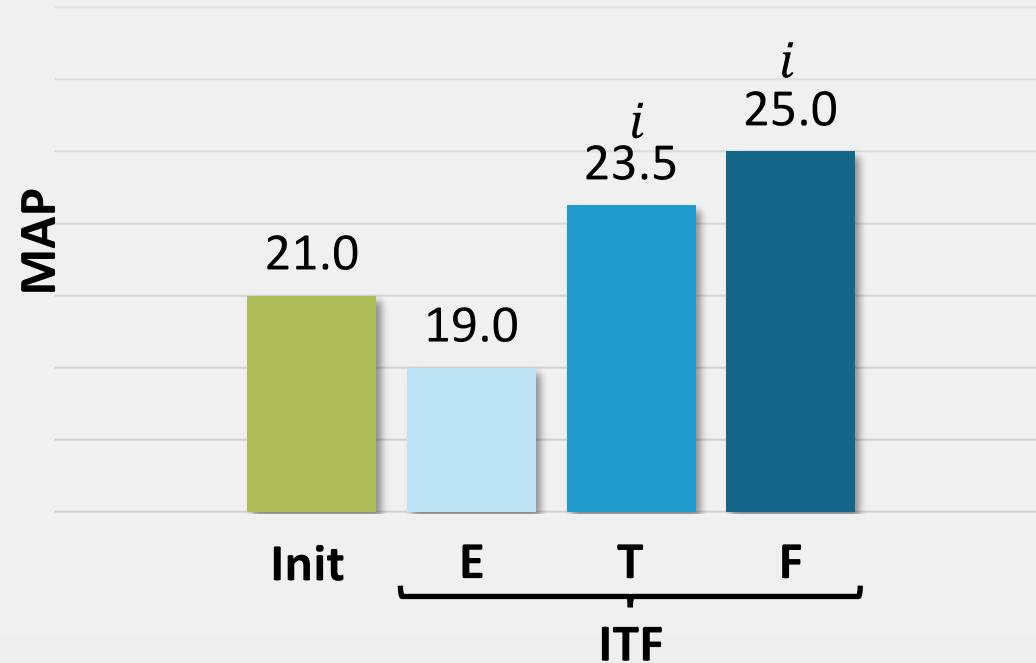
1. Entity relevance feedback is inferior to term feedback
2. Integrating term and entity feedback is of much merit

'*': statistically significant difference with T-X for $X \in \{RM_{\#3}, ITF\}$

'f': statistically significant difference with F-X for $X \in \{RM_{\#3}, ITF\}$

Init Vs. True Relevance Feedback

ClueWeb



Findings:

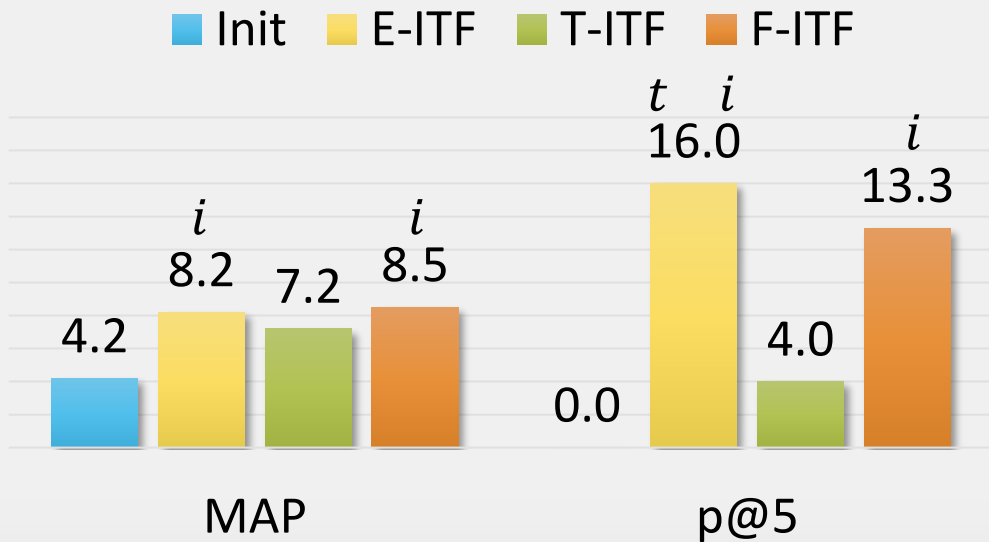
1. Term relevance feedback (T-ITF) outperforms Init
2. Entity relevance feedback (E-ITF) underperforms Init
3. F-ITF is our best performing method

'i': statistically significant difference with Init

Difficult Queries

- Queries with no relevant document in the top 10 in Init
 - ClueWeb: 45 difficult queries

ClueWeb



Finding:

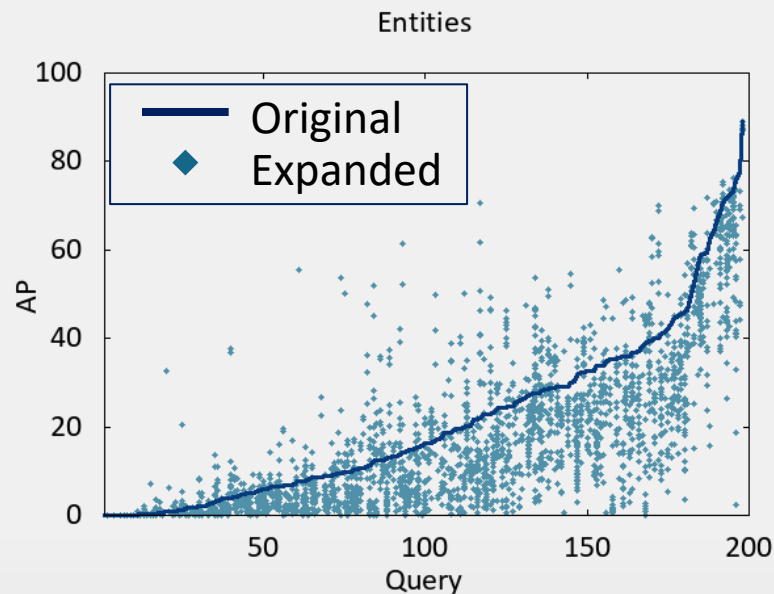
Entity relevance feedback is effective for difficult queries

'i': statistically significant difference with Init

't': statistically significant difference with T-ITF

Boosters Vs. Poison Pills

- Booster – a **relevant** token that **improves** performance when added alone to the query
- Poison pill – a **relevant** token that **degrades** performance when added alone to the query



	# topics	Tokens	% poison pills	% boosters
ClueWeb	198	Terms	73.5	26.5
		Entities	76.7	23.3

- Aligned with previous work on terms [Anick '03, Nemeth et al. '04, Ruthven '51, Tan et al. '07]

Summary

- The first usage of true entity relevance feedback for document retrieval
- We created the ETReF dataset
- Language model-based approaches that utilize entity or term relevance feedback
 - Entity relevance feedback is inferior to using term feedback
 - Entity relevance feedback is highly effective for difficult queries
- Integrating term and entity relevance feedback is highly effective